
COM-11102 Data Structures

Credits

8.0 (4 hrs lecture)

Lead Instructor

Silvia Guardati

Textbook

Java Software Structures – Designing and using Data Structures, John Lewis y Joseph Chase, Fourth edition, Pearson, 2013.

Supplemental materials

Thinking Recursively with Java, Eric Roberts, John Wiley & Sons, 2006.

Estructuras de Datos Básicas - Programación Orientada a Objetos con Java. Silvia Guardati. Alfaomega. 2015.

Estructuras de datos, Osvaldo Cairó y Silvia Guardati, McGraw-Hill, Tercera edición, 2006.

Course Description

In this course several linear data structures are covered. Students learn to design and implement these data structures using object-oriented programming techniques. Students also solve problems by applying these data structures and recursive programming.

Prerequisites

Algorithms and Programs (COM- 11101)

Major Status

Required

Course Outcomes

1. Use the basic concepts of object-oriented programming.
2. Describe how linear data structures works.
3. Successfully implement these data structures (with arrays and linked list) using interfaces and classes.
4. Use objects of these data structures to solve problems.
5. Apply recursive algorithms to solve problems.

Student Outcomes

- e) An ability to identify, formulate, and solve engineering problems.
- g) An ability to communicate effectively orally, with a wide diversity of audiences.

Course Topics

- Software quality. Analysis of algorithms: growth functions and BigO Notation.
- Selected topics in object-oriented programming: interfaces, inheritance, class hierarchies, polymorphism, method overloading, the static modifier, exceptions, wrapper classes.
- Queues and Stacks.
- Recursion.
- Set Collection.
- Linked Structures.
- List.